Appendix C: Search Descriptors

A Why a Descriptor Space

The genetic search never partitions candidates in their raw form. A candidate is a long per-layer vector: a bit-width for every weight matrix in the quantization search, a rank level for every matrix in the low-rank search, or a drop flag for every sub-block in the layer-drop search. For a model with dozens of layers this is on the order of a hundred dimensions, and clustering or stratifying directly in that space is hopeless, since covering each axis even once would take an impractical number of candidates and every candidate costs a full forward pass to evaluate.

Each candidate g is therefore mapped to a short, fixed-length **descriptor**: a six-number summary that captures the *style* of the allocation rather than its exact entries,

$$\Phi(g) = \big(\phi_1(g), \ldots, \phi_6(g)\big) \in [0,1]^6.$$

Two candidates that spend their budget the same way (early layers, attention-heavy, clustered, and so on) land near each other in Φ-space even if their individual entries differ. Every ϕ_k is normalised to $[0,1]$ and is read straight from the candidate vector, with no extra forward pass.

> **Role in the search.** The Centroidal Voronoi Tessellation of Appendix B is built in this descriptor space. Clustering there is cheap and meaningful, and it gives each island ownership of a structurally distinct region of allocation styles, so the populations stay diverse instead of collapsing onto one strategy.

The three searches use different descriptors because their candidates mean different things, but all six axes in each case follow the same intent: locate the budget along depth, split it between attention and feed forward, and measure how evenly or unevenly it is spread.

B Layer-Drop Descriptors

A layer-drop candidate marks attention and feed-forward sub-blocks for removal. Its descriptor records where the removals sit and how they are shaped.

- `drop_com`: the average depth of the dropped sub-blocks, scaled so that 0 means all drops are early and 1 means all are late. Writing $\mathcal{P}$ for the set of dropped depths,

$$\phi_{\text{com}} = \frac{1}{|\mathcal{P}|\,(L-1)} \sum_{p\in\mathcal{P}} p.$$

- `block_drop_share`: fraction of drops that remove a whole layer (both its attention and its feed forward), as opposed to only one side.
- `attn_only_share`: fraction of drops that remove only the attention sub-block.
- `mlp_only_share`: fraction of drops that remove only the feed-forward sub-block.
- `drop_clustering`: how bunched the drops are along depth, computed from the spread of the gaps between consecutive drops (1 = one contiguous run, 0 = evenly scattered).
- `total_drops_norm`: the number of dropped sub-blocks divided by the total available ($2L$ for L layers).

C Quantization Descriptors

A quantization candidate assigns a bit-width to each weight matrix. Its descriptor measures where the bits are concentrated and how precision is shared inside a block.

- `early_quarter_share`: fraction of the total bit budget spent in the first quarter of the layers.
- `late_quarter_share`: fraction spent in the last quarter of the layers.
- `qkv_share`: fraction of the bits spent on the query, key, and value input projections.
- `down_proj_share`: within the feed-forward block, the share of bits given to the down-projection rather than the up-projection.
- `longest_high_bit_run`: the longest run of consecutive blocks whose bit-width is above the candidate's own average, divided by the number of blocks.
- `numel_weighted_avg`: the parameter-weighted mean bit-width, normalised by the largest available level $b_{\max}$,

$$\phi_{\text{avg}} = \frac{\sum_i b_i\,|W_i|}{b_{\max}\sum_i |W_i|},$$

  a smooth stand-in for overall precision.

D Low-Rank and Recovery Descriptors

A low-rank candidate assigns a rank level to each matrix; the parameter cost of a layer grows with its rank. The descriptor reuses the depth and attention axes and adds two that come for free from the singular value spectrum.

- `early_rank_share`: fraction of the parameter budget placed in the first quarter of the layers.
- `late_rank_share`: fraction placed in the last quarter.
- `attn_rank_share`: fraction placed on the attention matrices rather than the feed-forward ones.
- `energy_mean`: the average fraction of each matrix's spectral energy that the chosen rank retains, read directly from the singular values σ_{ij},

$$\phi_{\text{energy}} = \frac{1}{K}\sum_{i=1}^{K} \frac{\sum_{j=1}^{r_i}\sigma_{ij}^2}{\sum_j \sigma_{ij}^2}.$$

- `energy_spread`: the standard deviation of that retained energy across layers, measuring how uneven the retention is.
- `longest_high_run`: the longest run of consecutive blocks whose rank level is above the candidate's average, divided by the number of blocks.

The rank-based recovery search reuses this same descriptor family, computed on the per-layer residual ranks and on the energy of the residual singular values, since a recovery candidate is itself a rank-allocation vector.